[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

[2]:
```
data = {
    "OrderID": [
        "1001", "1002", "1003", "1004", "1005",
        "1006", "1007", "1008", "1009", "1010",
        "1011", "1012", "1013", "1014", "1015",
        "1002", "1008", "1016", "1017", "1018"
    ],

    "OrderDate": [
        "2026-10-01", "2026-10-02", "2026-10-03",
        "2026-10-04", "2026-10-05", "2026-10-06",
        "2026-10-07", "2026-10-08", "2026-10-09",
        "2026-10-10", "2026-10-11", "2026-10-12",
        "invalid-date", "2026-10-14", "2026-10-15",
        "2026-10-02", "2026-10-08",
        "2026-10-16", "2026-10-17", "2026-10-18"
    ],

    "Product": [
        " Laptop ", "mouse", "KEYBOARD", "Monitor",
        "headphones", "MOUSE ", "Laptop", "keyboard",
        "MONITOR", "Mouse", "laptop", "Headphones",
        "Keyboard", "monitor", "MOUSE",
        "mouse", "keyboard", "Laptop", "Monitor", "Mouse"
    ],

    "Category": [
        "Electronics", "accessories", "ACCESSORIES",
        "electronics", "Accessories", "ACCESSORIES",
        "electronics", "Accessories", "ELECTRONICS",
        "accessories", "Electronics", "Accessories",
```

[3]:
```
df.head()
```

[3]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2	50000	Chennai
1	1002	2026-10-02	mouse	accessories	3	500	chennai
2	1003	2026-10-03	KEYBOARD	ACCESSORIES	NaN	1200	MADURAI
3	1004	2026-10-04	Monitor	electronics	1	15000	Chennai
4	1005	2026-10-05	headphones	Accessories	2	2500	Coimbatore

[4]:
```
df.tail()
```

[4]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
15	1002	2026-10-02	mouse	accessories	3	500	chennai
16	1008	2026-10-08	keyboard	Accessories	2	1200	madurai
17	1016	2026-10-16	Laptop	Electronics	1	500000	CHENNAI
18	1017	2026-10-17	Monitor	Electronics	2	200000	Madurai
19	1018	2026-10-18	Mouse	Accessories		500	NaN

[5]:
```
df.shape
```

[5]:
```
(20, 7)
```

[6]:
```
df.columns
```

[6]:
```
Index(['OrderID', 'OrderDate', 'Product', 'Category', 'Quantity', 'UnitPrice',
       'City'],
      dtype='str')
```

[6]:
```
df.columns
```

[6]:
```
Index(['OrderID', 'OrderDate', 'Product', 'Category', 'Quantity', 'UnitPrice',
       'City'],
      dtype='str')
```

[7]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderID    20 non-null     str
 1   OrderDate  20 non-null     str
 2   Product    20 non-null     str
 3   Category   20 non-null     str
 4   Quantity   19 non-null     str
 5   UnitPrice  19 non-null     str
 6   City       19 non-null     str
dtypes: str(7)
memory usage: 1.2 KB
```

[8]:
```
df.describe()
```

[8]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
count	20	20	20	20	19	19	19
unique	18	18	15	6	7	9	9
top	1002	2026-10-02	mouse	Accessories	2	500	chennai
freq	2	2	2	6	7	6	3

[10]:
```
df.isnull().sum()
```

[10]:
```
OrderID      0
OrderDate    0
Product      0
Category     0
Quantity     1
UnitPrice    1
City         1
dtype: int64
```

[11]:
```
df = df.replace(r"^\s*$", np.nan, regex=True)
df.isnull().sum()
```

[11]:
```
OrderID      0
OrderDate    0
Product      0
Category     0
Quantity     2
UnitPrice    1
City         1
dtype: int64
```

[13]:
```
df.duplicated().sum()
```

[13]:
```
np.int64(2)
```

[14]:
```
df[df.duplicated()]
```

[14]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
15	1002	2026-10-02	mouse	accessories	3	500	chennai
16	1008	2026-10-08	keyboard	Accessories	2	1200	madurai

[15]:
```
df = df.drop_duplicates()
df.duplicated().sum()
```

[15]:
```
np.int64(0)
```

[16]:
```
df.shape
```

[16]:
```
(18, 7)
```

[17]:
```
df["Product"] = df["Product"].str.strip()
df["Product"] = df["Product"].str.title()

df["Category"] = df["Category"].str.strip()
df["Category"] = df["Category"].str.title()

df["City"] = df["City"].str.strip()
df["City"] = df["City"].str.title()

df.head(10)
```

[17]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
0	1001	2026-10-01	Laptop	Electronics	2	50000	Chennai
1	1002	2026-10-02	Mouse	Accessories	3	500	Chennai
2	1003	2026-10-03	Keyboard	Accessories	NaN	1200	Madurai
3	1004	2026-10-04	Monitor	Electronics	1	15000	Chennai
4	1005	2026-10-05	Headphones	Accessories	2	2500	Coimbatore
5	1006	2026-10-06	Mouse	Accessories	4	500	Chennai
6	1007	2026-10-07	Laptop	Electronics	1	55000	Madurai
7	1008	2026-10-08	Keyboard	Accessories	2	1200	Madurai
8	1009	2026-10-09	Monitor	Electronics	1	NaN	Chennai
9	1010	2026-10-10	Mouse	Accessories	5	500	Coimbatore

[18]:
```
print(df["Product"].unique())
print(df["Category"].unique())
print(df["City"].unique())
```

```
<StringArray>
['Laptop', 'Mouse', 'Keyboard', 'Monitor', 'Headphones']
Length: 5, dtype: str
<StringArray>
['Electronics', 'Accessories']
Length: 2, dtype: str
<StringArray>
['Chennai', 'Madurai', 'Coimbatore', nan]
Length: 4, dtype: str
```

[19]:
```
df["OrderDate"] = pd.to_datetime(
    df["OrderDate"],
    errors="coerce"
)

df["OrderDate"].dtype
```

[19]:
```
dtype('<M8[us]')
```

[20]:
```
df[df["OrderDate"] > "2026-10-10"]
```

[20]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City
10	1011	2026-10-11	Laptop	Electronics	2	60000	Chennai
11	1012	2026-10-12	Headphones	Accessories	3	2500	Madurai
13	1014	2026-10-14	Monitor	Electronics	1	15000	Coimbatore
14	1015	2026-10-15	Mouse	Accessories	2	500	Madurai
17	1016	2026-10-16	Laptop	Electronics	1	500000	Chennai
18	1017	2026-10-17	Monitor	Electronics	2	200000	Madurai

[21]:
```
df["OrderID"] = pd.to_numeric(df["OrderID"], errors="coerce")
df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["UnitPrice"] = pd.to_numeric(df["UnitPrice"], errors="coerce")

df.info()
```

```
<class 'pandas.DataFrame'>
Index: 18 entries, 0 to 19
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   OrderID    18 non-null     int64
 1   OrderDate  17 non-null     datetime64[us]
 2   Product    18 non-null     str
 3   Category   18 non-null     str
 4   Quantity   15 non-null     float64
 5   UnitPrice  17 non-null     float64
 6   City       17 non-null     str
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 1.1 KB
```

[22]:
```
df.isnull().sum()
```

[22]:
```
OrderID      0
OrderDate    1
Product      0
Category     0
Quantity     3
UnitPrice    1
City         1
dtype: int64
```

[23]:
```
df["Quantity"] = df["Quantity"].fillna(0)
```

[25]:
```
df["City"] = df["City"].fillna("Unknown")
```

[26]:
```
df = df.dropna(subset=["OrderDate"]).copy()
df.isnull().sum()
```

[26]:
```
OrderID      0
OrderDate    0
Product      0
Category     0
Quantity     0
UnitPrice    0
City         0
dtype: int64
```

[27]:
```
Q1 = df["UnitPrice"].quantile(0.25)
Q3 = df["UnitPrice"].quantile(0.75)

IQR = Q3 - Q1

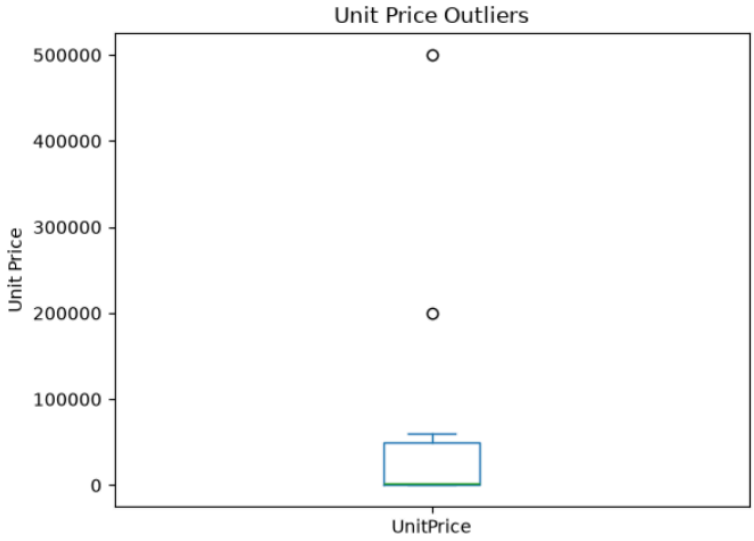
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)
```

```
Q1: 500.0
Q3: 50000.0
IQR: 49500.0
Lower Limit: -73750.0
Upper Limit: 124250.0
```

[29]:
```
df["UnitPrice"].plot(
    kind="box",
    title="Unit Price Outliers"
)

plt.ylabel("Unit Price")
plt.show()
```



[30]:
```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
df.head()
```

[30]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City	Revenue
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai	100000.0
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	Chennai	1500.0
2	1003	2026-10-03	Keyboard	Accessories	0.0	1200.0	Madurai	0.0
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai	15000.0
4	1005	2026-10-05	Headphones	Accessories	2.0	2500.0	Coimbatore	5000.0

[31]:
```
print("Total Revenue:", df["Revenue"].sum())
print("Average Revenue:", df["Revenue"].mean())
print("Maximum Revenue:", df["Revenue"].max())
print("Minimum Revenue:", df["Revenue"].min())
```

```
Total Revenue: 1229400.0
Average Revenue: 72317.64705882352
Maximum Revenue: 500000.0
Minimum Revenue: 0.0
```

[32]:
```
def price_category_function(price):
    if price < 1000:
        return "Low"
    elif price < 20000:
        return "Medium"
    else:
        return "High"
```

[33]:
```
df["PriceCategory"] = df["UnitPrice"].apply(
    price_category_function
)
df.head(10)
```

[33]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City	Revenue	PriceCategory
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai	100000.0	High
1	1002	2026-10-02	Mouse	Accessories	3.0	500.0	Chennai	1500.0	Low
2	1003	2026-10-03	Keyboard	Accessories	0.0	1200.0	Madurai	0.0	Medium
3	1004	2026-10-04	Monitor	Electronics	1.0	15000.0	Chennai	15000.0	Medium
4	1005	2026-10-05	Headphones	Accessories	2.0	2500.0	Coimbatore	5000.0	Medium
5	1006	2026-10-06	Mouse	Accessories	4.0	500.0	Chennai	2000.0	Low
6	1007	2026-10-07	Laptop	Electronics	1.0	55000.0	Madurai	55000.0	High
7	1008	2026-10-08	Keyboard	Accessories	2.0	1200.0	Madurai	2400.0	Medium
8	1009	2026-10-09	Monitor	Electronics	1.0	2500.0	Chennai	2500.0	Medium
9	1010	2026-10-10	Mouse	Accessories	5.0	500.0	Coimbatore	2500.0	Low

[34]:
```
df["PriceCategory"].value_counts()
```

[34]:
```
PriceCategory
Medium    7
High      5
Low       5
Name: count, dtype: int64
```

[35]:
```
print("Shape:", df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nData Types:")
df.info()

print("\nStatistical Summary:")
display(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicates:")
print(df.duplicated().sum())
```

```
Shape: (17, 9)

Column Names:
Index(['OrderID', 'OrderDate', 'Product', 'Category', 'Quantity', 'UnitPrice',
       'City', 'Revenue', 'PriceCategory'],
      dtype='str')

Data Types:
<class 'pandas.DataFrame'>
Index: 17 entries, 0 to 19
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   OrderID        17 non-null     int64
 1   OrderDate      17 non-null     datetime64[us]
 2   Product        17 non-null     str
 3   Category       17 non-null     str
 4   Quantity       17 non-null     float64
 5   UnitPrice      17 non-null     float64
 6   City           17 non-null     str
```

[36]:
```
df.groupby("City")["Revenue"].sum()
```

[36]:
```
City
Chennai       741000.0
Coimbatore     22500.0
Madurai       465900.0
Unknown            0.0
Name: Revenue, dtype: float64
```

[37]:
```
df.groupby("Product")["Revenue"].sum()
```

[37]:
```
Product
Headphones     12500.0
Keyboard        2400.0
Laptop        775000.0
Monitor       432500.0
Mouse           7000.0
Name: Revenue, dtype: float64
```

[38]:
```
df[df["Revenue"] > 50000]
```

[38]:

	OrderID	OrderDate	Product	Category	Quantity	UnitPrice	City	Revenue	PriceCategory
0	1001	2026-10-01	Laptop	Electronics	2.0	50000.0	Chennai	100000.0	High
6	1007	2026-10-07	Laptop	Electronics	1.0	55000.0	Madurai	55000.0	High
10	1011	2026-10-11	Laptop	Electronics	2.0	60000.0	Chennai	120000.0	High
17	1016	2026-10-16	Laptop	Electronics	1.0	500000.0	Chennai	500000.0	High
18	1017	2026-10-17	Monitor	Electronics	2.0	200000.0	Madurai	400000.0	High

[39]:
```
df.groupby("Product")["Revenue"].sum().plot(
    kind="bar",
    title="Revenue by Product"
)

plt.xlabel("Product")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```

